Unveiling the electrodynamic nature of spacetime collisions
Supplemental Material

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Appendix A: Tetrads for use with numerical relativity simulations

In order to analyze the dynamical spacetime of a binary black hole collision, we cannot make the same simplifying assumptions we have taken in the context of a Kerr metric, where special choices of tetrads readily exist [1]. Instead, we need to operate within the framework of a dynamically evolved metric, the evolution of which imposes its own time slicing. The split of the spacetime metric, $g_{\mu\nu}$ follows the standard Arnowitt-Deser-Misner (ADM) equations [2], where the future-directed unit normal vector of spacelike hypersurfaces are

$$n_\mu = (-\alpha, 0), \qquad n^\mu = \frac{1}{\alpha}(1, -\beta^i), \tag{A1}$$

where α is the lapse function and β^i is the shift vector. The line element is given by

$$\mathrm{d}s^2 = g_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu = -\alpha^2\mathrm{d}t^2 + \gamma_{ij}(\mathrm{d}x^i + \beta^i \mathrm{d}t)(\mathrm{d}x^j + \beta^j \mathrm{d}t), \tag{A2}$$

where γ_{ij} is the induced spatial metric, whose determinant is related to spacetime metric as $\sqrt{-g} = \alpha\sqrt{\gamma}$.

In order to naturally take this into account, we choose to align the time coordinate with the normal observer n^μ, i.e.,

$$A^\mu_{(\hat{t})} = n^\mu = (\alpha^{-1}, -\alpha^{-1}\beta^i). \tag{A3}$$

In terms of a spatial observer, we follow an approach taken for solving local hydrodynamics problems [3, 4], in particular we adopt the tetrad presented in Ref. [5], which we summarize in the following. As for the spatial alignment, the system naturally has preferred directions of motion, namely radially decaying orbital motion in a plane perpendicular to the $z-$coordinate direction. We therefore choose the $\hat{z}$ tetrad to align with the orbital angular momentum, i.e.,

$${}^{(3)}A^{(\hat{z})}_\mu \propto (0, 0, 1) \tag{A4}$$

After some algebra, we can obtain

$$A^\mu_{(\hat{z})} = \frac{g^{03}g^{0\mu} - g^{00}g^{3\mu}}{(g^{00}(g^{00}g^{33} - g^{03}g^{03}))^{1/2}}, \tag{A5}$$

$$= \left(0, \frac{\gamma^{zi}}{\alpha\sqrt{\gamma^{zz}}}\right) \tag{A6}$$

Finally, we can go through the same procedure by aligning the $\hat{y}$ tetrad so that it is orthogonal to surfaces of constant coordinate y after projection onto the two-dimensional surface with constant t and z. The ultimate expression for the remaining two tetrads is

$$A^\mu_{(\hat{x})} = (0, (g_{11})^{-1/2}, 0, 0) \tag{A7}$$

$$= \left(0, \frac{1}{\sqrt{\gamma_{xx}}}, 0, 0\right) \tag{A8}$$

$$A^\mu_{(\hat{y})} = \frac{1}{(g_{11}(g_{11}g_{22} - g_{12}g_{12}))^{1/2}}(0, -g_{12}, g_{11}, 0) \tag{A9}$$

$$= \left(0, -\frac{\gamma_{xy}}{\sqrt{\gamma\gamma_{xx}\gamma^{zz}}}, \frac{\gamma_{xx}}{\sqrt{\gamma\gamma_{xx}\gamma^{zz}}}, 0\right) \tag{A10}$$

In a block-matrix form we can write it as

$$A^\mu{}_{\hat{\nu}} = e^\mu_{(\hat{\nu})} = \begin{pmatrix} 1/\alpha & 0 \\ -\beta^i/\alpha & A^i{}_{\hat{i}} \end{pmatrix}, \tag{A11}$$

where the spatial matrix $A^i{}_{\hat{i}}$ is an upper triangular matrix with components specified in eqs. eqs. (A6), (A7) and (A9) and satisfies

$$\beta^{\hat{i}} = A^{\hat{i}}{}_i \beta^i, \tag{A12}$$

$$\delta_{\hat{i}\hat{j}} A^{\hat{i}}{}_i A^{\hat{j}}{}_j = \gamma_{ij}. \tag{A13}$$

We can also invert the matrix eq. (A11) such that

$$A^{\hat{\mu}}{}_\nu = \theta^{(\hat{\mu})}_\nu = \begin{pmatrix} \alpha & 0 \\ \beta^{\hat{i}} & A^{\hat{i}}{}_i \end{pmatrix} \tag{A14}$$

White et al. [5] gives the components of inverse matrix in terms of the spacetime metric $g_{\mu\nu}$ and its inverse $g^{\mu\nu}$ (eq. (71) in their paper). However, since the field strength tensor are calculated via partial derivatives of the tetrad in eq. (7) in the main text, it's more convenient to rewrite the components in terms of solely covariant spacetime metric, simply because those are the fundamental evolved variables in the simulation. To this end, we show that the tetrad basis one-form can be written as

$$A^{\hat{0}}{}_\nu = \theta^{(\hat{0})}_\nu = -n_\nu, \tag{A15}$$

$$A^{\hat{1}}{}_\nu = \theta^{(\hat{1})}_\nu = A g_{1\nu}, \tag{A16}$$

$$A^{\hat{2}}{}_\nu = \theta^{(\hat{2})}_\nu = AB(g_{11}g_{2\nu} - g_{12}g_{1\nu}), \tag{A17}$$

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$$A^{\hat{3}}{}_{\nu} = \theta^{(\hat{3})}_{\nu} = A((g_{11}g_{3\nu} - g_{13}g_{1\nu}) - (g_{11}g_{2\nu} - g_{12}g_{1\nu})B^2C)^{1/2}. \tag{A18}$$

where the coefficients are

$$A = (g_{11})^{-1/2}, \tag{A19}$$

$$B = (g_{11}g_{22} - g_{12}g_{12})^{-1/2}, \tag{A20}$$

$$C = g_{11}g_{23} - g_{12}g_{13}. \tag{A21}$$

One can show that $A^{\hat{2}}{}_1 = A^{\hat{3}}{}_1 = A^{\hat{3}}{}_2 = 0$, i.e. the spatial part of the inverse tetrad is also an upper triangular matrix.

Appendix B: Gravitational magnetic fields of a non-rotating black hole

To calculate the magnetic field and compare it directly with the dipole field, we rotate the tetrad so that it aligns with the Cartesian coordinates. This is equivalent to multiply the diagonal tetrad defined in the black hole electrostatics section with a rotation matrix. The full expression of this tetrad is shown in eq. (37) in Ref. [1]. By adding this rotation, we construct a set of observers who are static in the spacetime and whose tetrads are parallel to each other. In this tetrad basis, the non-vanishing components of the gravitational magnetic field of a Schwarzschild black hole in Schwarzschild metric are:

$$B^{\hat{x}\theta} = -\frac{\left(-1+\sqrt{1-\frac{2M}{R}}\right)\sin\phi}{R^2} \simeq \frac{M\sin\phi}{R^3} + \mathcal{O}\left(\frac{1}{R^4}\right), \tag{B1}$$

$$B^{\hat{x}\phi} = -\frac{\left(-1+\sqrt{1-\frac{2M}{R}}\right)\cos\phi}{\tan\theta\, R^2} \simeq \frac{M\cos\phi\cot\theta}{R^3} + \mathcal{O}\left(\frac{1}{R^4}\right), \tag{B2}$$

$$B^{\hat{y}\theta} = -\frac{\left(-1+\sqrt{1-\frac{2M}{R}}\right)\cos\phi}{R^2} \simeq \frac{M\cos\phi}{R^3} + \mathcal{O}\left(\frac{1}{R^4}\right), \tag{B3}$$

$$B^{\hat{y}\phi} = -\frac{\left(-1+\sqrt{1-\frac{2M}{R}}\right)\sin\phi}{\tan\theta\, R^2} \simeq \frac{M\sin\phi\cot\theta}{R^3} + \mathcal{O}\left(\frac{1}{R^4}\right), \tag{B4}$$

$$B^{\hat{z}\phi} = \frac{\left(-1+\sqrt{1-\frac{2M}{R}}\right)}{R^2} \simeq -\frac{M}{R^3} + \mathcal{O}\left(\frac{1}{R^4}\right), \tag{B5}$$

where we expand the expression in the far field region, i.e. $R\to\infty$, and keep only the leading term $\propto R^{-3}$

Appendix C: Gravitational electromagnetic far-fields of a Kerr black hole

Assuming a Boyer-Lindquist form of the Kerr, the gravitational electric and magnetic fields can be expressed as follows in the far field limit. We are adopting the tetrad given in Ref. [1]. In the far field, the leading term of non-vanishing components of the gravitational electric field is

$$E^{\hat{t}R} \simeq \frac{M}{R^2} + \frac{\chi^2M^3(1-4\cos^2\theta)}{R^4} + \mathcal{O}\left(\frac{1}{R^5}\right), \tag{C1}$$

$$E^{\hat{x}R} \simeq \frac{2\chi M^3\sin\theta\sin\phi}{R^4} + \mathcal{O}\left(\frac{1}{R^5}\right), \tag{C2}$$

$$E^{\hat{y}R} \simeq -\frac{2\chi M^3\sin\theta\cos\phi}{R^4} + \mathcal{O}\left(\frac{1}{R^5}\right), \tag{C3}$$

and other components of the electric field are in order of $\mathcal{O}\left(\frac{1}{R^5}\right)$. Therefore, the electric field of a Kerr black hole consists of a dominant monopole field and a disturbed quadrupole field because of the deformed shape due to rotation. Meanwhile, the gravitational magnetic field takes the following form in the far-field limit:

$$B^{\hat{\alpha}\mu} \simeq \begin{pmatrix} 0 & -\frac{4\chi M^2\cos(\theta)}{R^3} & 0 & 0 \\ 0 & -\frac{3\chi M^2\sin\theta\cos\theta\sin\phi}{2R^3} & \frac{M\sin\phi}{R^3} & \frac{M\cot\theta\cos\phi}{R^3} \\ 0 & \frac{3\chi M^2\sin\theta\cos\theta\cos\phi}{2R^3} & \frac{M\cos\phi}{R^3} & \frac{M\cot\theta\sin\phi}{R^3} \\ 0 & 0 & 0 & -\frac{M}{R^3} \end{pmatrix} + \mathcal{O}\left(\frac{1}{R^4}\right), \tag{C4}$$

where $\hat{\alpha}$ runs through $\{\hat{t}, \hat{x}, \hat{y}, \hat{z}\}$ and μ runs through Boyer-Lindquist coordinates $\{t, r, \theta, \phi\}$. The result shows an asymptotic dipole decay and a quadrupole correction. The analytical study matches the numerical scaling behavior in the far field region, as shown in Fig. 2 in the main text.

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